

From Independent Scientist to Novel Performer

A Small-Business Operating, Milestone, and Capitalization Plan for a Phase 1 LLM-Advised Robotic Whipple (ChemicalQDevice)

Draft 1.0

Repository v4.5.0

20 figures

12 milestones

SBIR Phase I
\$306,000

9 months, 5 milestones

SBIR Phase II
\$1,300,000

24 months, 7 milestones

Five-year programme
\$3,500,000

60 months, direct

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San Diego | August 11, 2026

Independent research paper and practical adoption guide. It is not medical or regulatory advice and is not endorsed by the FDA, NIH, HHS, an IRB, ICH, or any sponsor. All figures derive from the author's repository sources and are illustrative unless tied to a cited reference.

Disclaimer: This work is independent and is not endorsed or sponsored by any trial sponsor, CRO, site, IRB, regulator, or medical society; and was adapted using Claude Code Opus 5.

The paper (v1.0) is at <https://doi.org/10.5281/zenodo.xxxxxxxx> and the repository (v4.5.0) is at <https://github.com/kevinkawchak/physical-ai-oncology-trials>. The paper files are deposited in funding/capitalization-plan.

Keywords Small business innovation research; Novel performers; Capitalization plan; Milestone schedule; Indirect cost; 21 CFR part 54; Pancreatic ductal adenocarcinoma; Robotic pancreaticoduodenectomy; Daraxonrasib; Physical AI

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Abstract

[DRAFTING INSTRUCTION] Write 200 to 240 words. Open with the conversion: the author has filed ten funding applications as an independent scientist, and one of the ten, application 05, is addressed to the only mechanism in the set built for a company. State the three numbers that govern the paper, \$306,000, \$1,300,000 and \$3,500,000, and the two derived ones, \$1,396,000 of direct work inside the award and \$2,104,000 outside it. Name the clause: not the report's Pioneer language and not its NOVEL PERFORMERS chapter, but the SBIR clause, which appears in Chapters I, III and IV. Close on what is being offered, which is a costed twelve-milestone schedule with an evidence artifact and a stop condition on every row. Source: `funding/pdac-funding-applications/final-apply/sections/sec-00-front.tex` for the abstract form, and `funding/science-golden-age/chunk-03-chapter-two-revitalizing-science-and-technology-enterprise.md` for the clause.

Executive Summary

[DRAFTING INSTRUCTION] Six short paragraphs, one per outline item A to E plus one on what the conversion costs. Each must carry at least one number. Do not restate the abstract. Sources: `funding/pdac-funding-applications/final-apply/sections/sec-08-budget-and-leverage.tex` for the budget frame, `funding/pdac-funding-applications/applications/app-05-nih-sbir-seed/` for the award split, and `funding/science-golden-age/chunk-08-annex-fiscal-year-2028-research-and-development-budget-priorities.md` for the 3 to 1 leverage target.

How to Read This Document

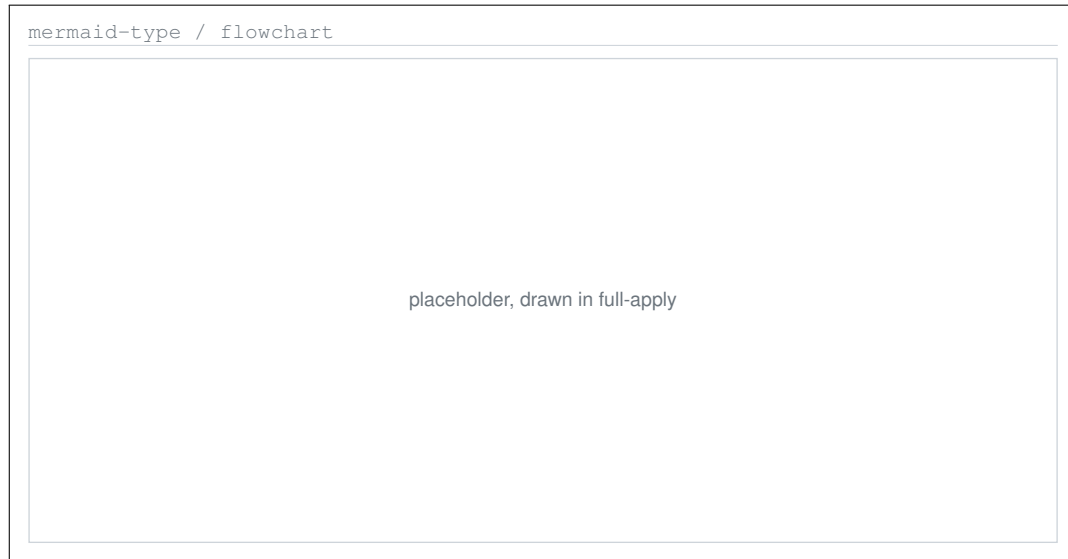
[DRAFTING INSTRUCTION] One short paragraph plus a four-row table mapping four readers to the sections each should read first: an SBIR program director, an ARPA-H programme manager, a private investor, and a UC San Diego intake committee. Source: `funding/pdac-funding-applications/final-apply/sections/sec-02-ten-applications.tex` for the reader-mapping form.

Fig	§	Platform	Source directory	The question it answers
1	1	Mermaid	mermaid/	Which clause does a company with no indirect-cost base qualify under?
2	1	D2	d2/	What kind of institution is a 2.6 FTE firm, on the report's own axes?
3	1	Graphviz	graphviz/	What does the same direct work cost under three overhead regimes?
4	2	Diagrams	diagrams-pyhton/	What is the shape of what the company holds today?
5	2	D2	d2/	What is owned, licensed, contracted, and absent, with terms?
6	2	PlantUML	plantuml/	What may the sponsor do alone, and what needs an institution?
7	3	Mermaid	mermaid/	What must hold at month nine for Phase II to begin?
8	3	D2	d2/	What does the same programme cost under two mechanisms?
9	3	Graphviz	graphviz/	Which work packages does the money reach, and which not?
10	4	PlantUML	plantuml/	Which capital positions may be held while a participant is on study?
11	4	D2	d2/	Where does every dollar come from, and what separates the tiers?
12	4	Mermaid	mermaid/	In what order is a private round executed during enrollment?
13	5	Mermaid	mermaid/	When does each milestone open, close, and produce its artifact?
14	5	Graphviz	graphviz/	What has to fail, and in what combination, to stop the programme?
15	5	PlantUML	plantuml/	What do three parties do at once to close one milestone?
16	6	D2	d2/	Which published quantities can a funder check before committing?
17	6	Graphviz	graphviz/	Where does each published number land in a filing?
18	7	Diagrams	diagrams-pyhton/	Who is employed, who is contracted, who is contributed?
19	8	Mermaid	mermaid/	What did the July and August 2026 contacts actually unlock?
20	10	Diagrams	diagrams-pyhton/	What survives if the programme stops, and who holds it?

The twenty figures, their section, their platform, the directory that carries each specification, and the one question each answers. Five mermaid, three plantuml, five d2, three diagrams, four graphviz, by purpose not quota.

Tab	§	Contents	The question it answers
1	0	Figure index, twenty rows	Where is each figure and what does it answer?
2	0	Table index, nineteen rows	Where is each table and what does it answer?
3	0	Four readers, four entry points	Which section should this reader open first?
4	1	Three clauses against four eligibility tests	Which clause survives the filter, and why?
5	1	Three overhead regimes on one direct base	What is the premium for routing through an institution?
6	2	The owned register, thirteen rows with DOIs	What exists today and where is it deposited?
7	2	The absent register, seven rows with signatories	What is missing and who alone can supply it?
8	3	The four-layer budget under two prices	What does the SBIR route reach, and what does it not?
9	3	Phase I line items, eight rows to \$306,000	What does nine months of Phase I actually buy?
10	3	Phase II line items, ten rows to \$1,300,000	What does twenty-four months of Phase II buy?
11	4	The §54.2 and §54.4 disclosure triggers	What must be disclosed, at what threshold, on which form?
12	4	The capitalization table across five stages	Who owns what, and does the SBIR test still pass?
13	5	The twelve milestones, costed and gated	What is the evidence, the cost, and the stop condition?
14	5	Milestone artifacts against funder mechanism	Which mechanism pays for which milestone?
15	6	Six checkable quantities with limitations	What can a reviewer verify against a published source?
16	6	Cost and schedule against industry benchmarks	What did the prior work cost, and what does it cost elsewhere?
17	7	The 2.6 FTE by employment status	Who is on the payroll and who is not?
18	9	Risks, likelihood, and the mitigation held	What could go wrong and what is already in place?
19	11	Abbreviations, two column pairs	What does this term mean in this paper?

The nineteen tables, their section, their contents, and the one question each answers. Every table is set to the full body measure with ragged-right fixed columns, so no cell shows a stretched interword gap.



Three clauses of one report, four eligibility tests, and the only clause a San Diego firm with no indirect-cost base survives. The two that fail, fail on size, not on merit or on subject matter.

1 The Novel-Performer Case

1.1 Three Clauses, One Applicant

[DRAFTING INSTRUCTION] *Open by naming the three candidate clauses and quoting each in one sentence. Clause 1 is the individual-scientist language of Chapter I, which the ten prior applications were written under. Clause 2 is the NOVEL PERFORMERS section of Chapter II. Clause 3 is the SBIR clause, which appears three times: in the Chapter I summary of recommendations, in Chapter III on deploying SBIR and STTR strategically to couple federally-seeded companies with the scientific enterprise, and in Chapter IV on SBIR opening doors for technician-founded ventures. Quote each from `funding/science-golden-age/chunk-01-front-matter-and-summary.md`, `funding/science-golden-age/chunk-03-chapter-two-revitalizing-science-and-technology-enterprise.md`, `funding/science-golden-age/chunk-04-chapter-three-securing-dominance-in-critical-and-emerging-technologies.md` and `funding/science-golden-age/chunk-05-chapter-four-science-and-technology-better-lives-of-all-americans.md`. Cite `kratsios2026sciencegoldenage`.*

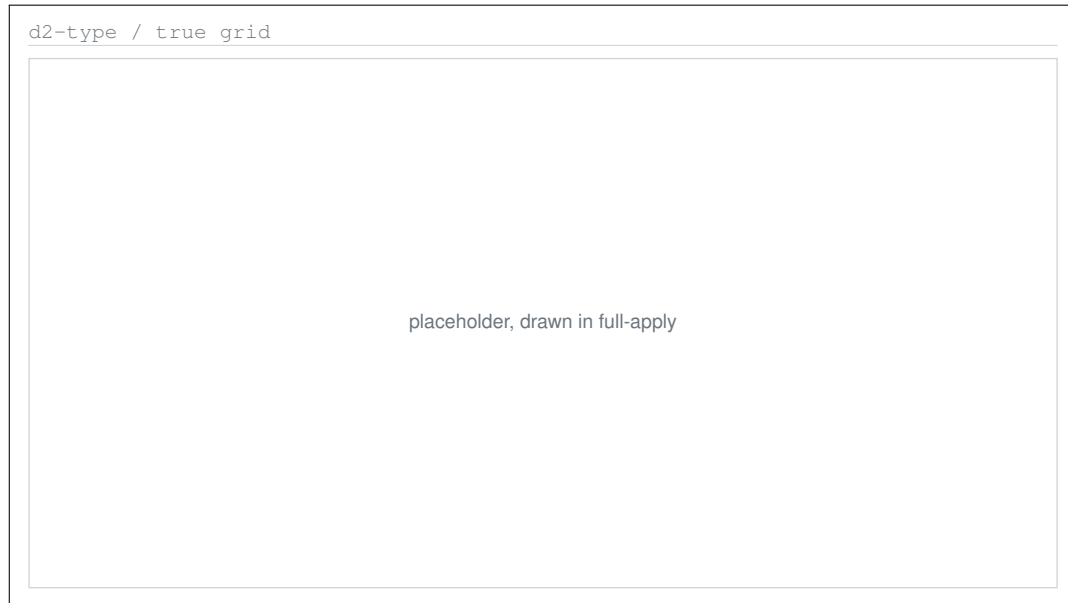
[DRAFTING INSTRUCTION] *After Figure 1, set Table 4: the three clauses against the four eligibility tests T1 to T4, with the ChemicalQDevice answer in the last column. The four tests and their report basis are given in `funding/capitalization-plan/mermaid/fig-01-clause-eligibility-filter.md`. Use five columns: test, question, report basis, this company, verdict.*

1.2 Why the Pioneer Language Does Not Fit an Entity

[DRAFTING INSTRUCTION] *Two paragraphs. The first states what changed: the applicant is now a California corporation and not a person, so a person-based mechanism has no subject to fund. The second is the honest concession, which must not be softened: the individual-scientist framing is what made the ten prior applications distinctive and is what the report actually rewards. Name the cost and forward-reference §9. Source: `funding/pdac-funding-applications/final-apply/sections/sec-01-golden-age-mandate.tex`.*

1.3 Why the Novel-Performers Chapter Does Not Fit Either

[DRAFTING INSTRUCTION] *One paragraph with three numbers taken from the chapter itself: tens of millions of dollars, coordinated teams of ten to a hundred people, and timelines of half a decade. Set them beside this company's \$1,606,000, 2.6 FTE and 33 months. The conclusion is that the chapter describes a real gap that this company does*



The report's own seven activities, its four institutional forms, and a fifth column it does not carry. An SBIR firm of 2.6 people is well suited to two of the seven and unsuited to two others.

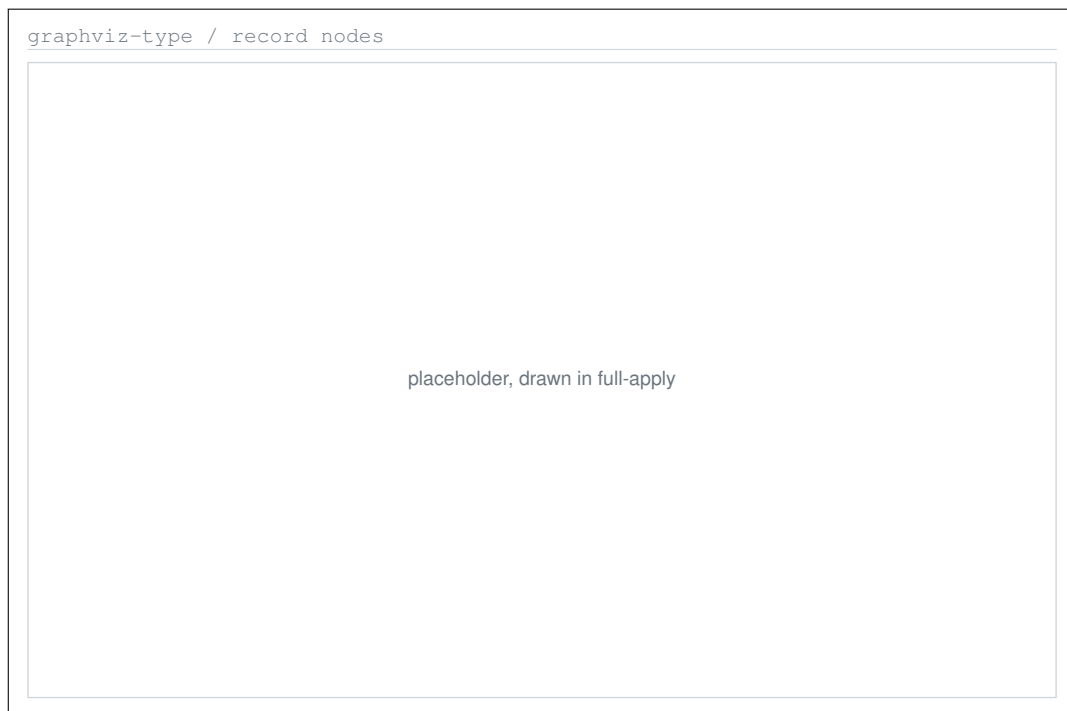
*not fill, and that claiming otherwise would be the kind of overreach the report is written against. Source: `funding/science-golden-age/chunk-03-chapter-two-revitalizing-science-and-technology-ent`
`erprise.md`, the NOVEL PERFORMERS section.*

[DRAFTING INSTRUCTION] After Figure 2, one paragraph reading the fifth column: two well suited, three partly suited, two less suited, and the observation that a firm claiming to be well suited to all seven would be claiming to be a university, a corporate lab, a federal lab and an FRO at once. The row-by-row reasoning is in `funding/capitalization-plan/d2/fig-02-institutional-form-grid.d2.md`.

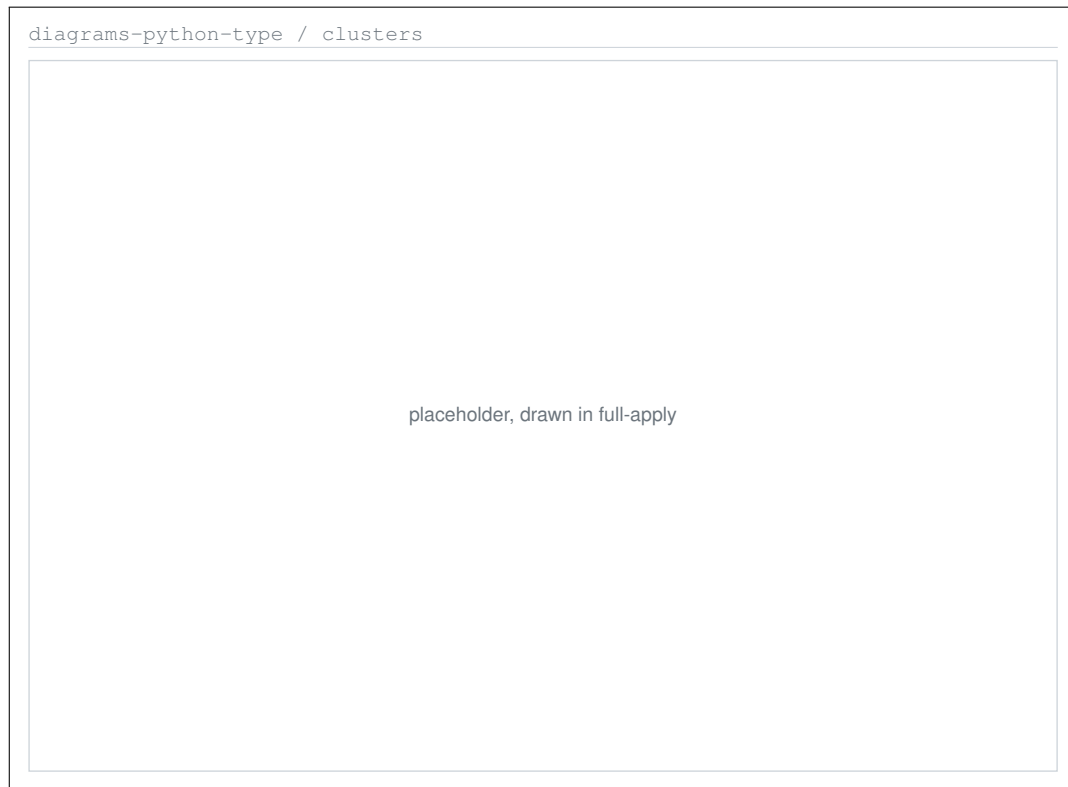
1.4 The Clause That Fits, and What It Is Worth

[DRAFTING INSTRUCTION] Open with the structural fact: this company has no negotiated indirect-cost rate agreement, no facilities to recover, and no administrative layer to fund. Then price it. Figure 3 and Table 5 carry the arithmetic from `funding/capitalization-plan/graphviz/fig-03-indirect-cost-decomposition.gv.md`. Cite `nihsbirseed` for the 40 percent allowance and `ecfr2026uniformguidance` for the *de minimis* rate.

[DRAFTING INSTRUCTION] Set Table 5 immediately after Figure 3: seven rows, four columns (line, route A university, route B full allowance, route C this plan), ending in the premium row. Every figure must reconcile; the checks are listed in `funding/capitalization-plan/graphviz/README.md`. Close the section with two sentences on what the 33.1 percent premium means to a program officer who has a fixed appropriation.



The same 1,396,000 of direct work priced three ways. A university F and A rate at 57 percent adds 741,000, the full SBIR allowance adds 695,000, and the rate this plan claims adds 210,000 in all.



Four zones and thirteen tiles. The owned zone holds everything the company made; the contracted zone is drawn at full size and holds nothing, because an empty zone is the finding this figure reports.

2 The Entity and the Asset

2.1 What the Entity Is

[DRAFTING INSTRUCTION] *Four sentences and no more: ChemicalQDevice, a California corporation, sole owner and sole officer Kevin Kawchak, San Diego, no employees at the date of writing, no negotiated indirect-cost rate agreement, no facilities. State that the company is more than 50 percent owned and controlled by one United States individual, which is the eligibility test at ecfr2026sbiownership, and that §4 is written to keep it so.*

2.2 Owned

[DRAFTING INSTRUCTION] *Set Table 6 from the thirteen owned rows in funding/capitalization-plan/d2/fig-05-asset-register-records.d2.md, each with its instrument, date and DOI. Every DOI must print as text and link, using \dlink. Then two paragraphs: what the thirteen rows cost to produce, drawing the estimate from funding/pdac-funding-applications/final-apply/sections/sec-05-trial-evidence.tex, and what a funder is therefore not paying for. Cite foundingdocs, onprewhipl, phase2proto, phaselind, qspmetpancre and tenapps2026.*

2.3 Licensed, and Contracted

[DRAFTING INSTRUCTION] *One short paragraph on the three licensed rows, none exclusive, and one on the contracted table, which has no rows. State plainly that no clinical trial agreement, drug supply agreement, letter of authorization, confidentiality agreement, subaward or trial-specific insurance binder exists. Source: funding/potential-partners/UC-San-Diego/ and funding/pdac-funding-applications/final-apply/sections/sec-07-moores-partnership.tex.*

d2-type / sql tables

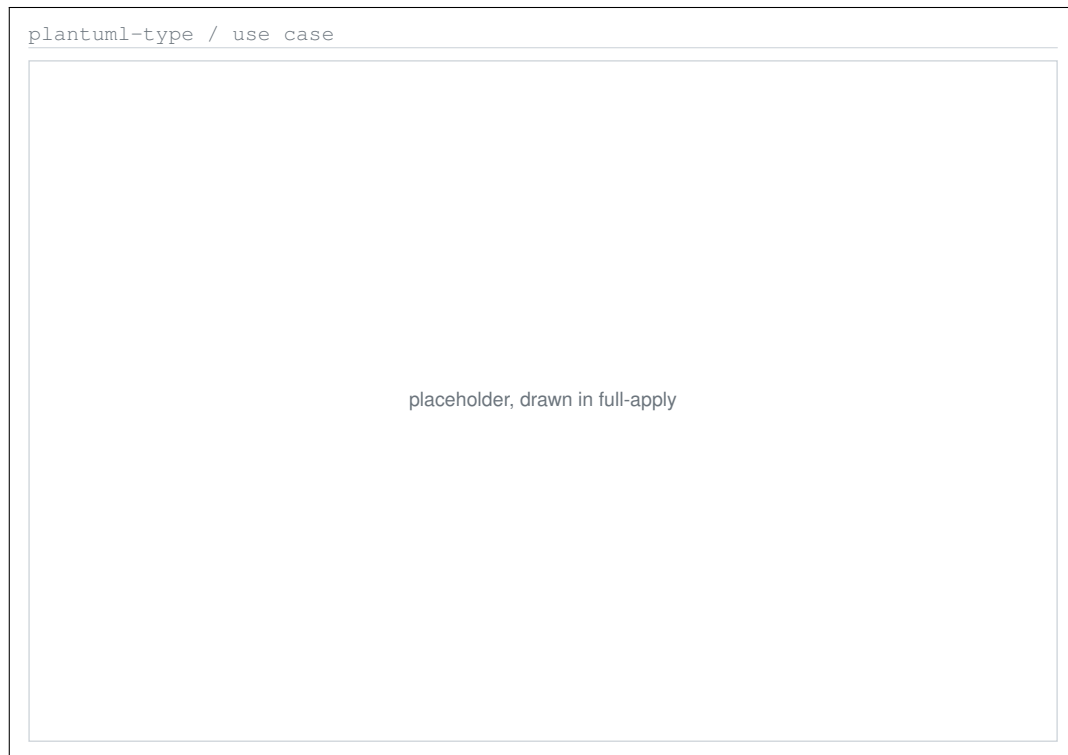
placeholder, drawn in full-apply

The asset register as four typed records. Thirteen rows are owned outright, three are licensed on public terms, seven are absent, and the contracted table has no rows at all as of August 2026.

2.4 Absent

[DRAFTING INSTRUCTION] Set Table 7: the seven absent rows, each with the milestone it blocks and the single party that can sign it. The point of the column headed "company can obtain alone" is that it reads false on every row. Source: `funding/capitalization-plan/d2/fig-05-asset-register-records.d2.md`.

[DRAFTING INSTRUCTION] After Figure 6, one paragraph on the corridor: three use cases sit in it, each blocked by exactly one signature, and no amount of company work removes any of the three. Then a short closing paragraph dating the register: on 13 January 2026 the owned table held two rows and the absent table held the same seven, so the seven have been unchanged for seven months while the owned table grew by eleven. Source: `funding/supplementary/source-files/Physical-AI-Oncology-Trial-Competition-Proposal.zip` and `funding/supplementary/PhysicalAIOncologyTrialFoundingDocuments.md`. Cite `paiproposal`.



Two system boundaries, twelve use cases, and the corridor between them holding three that neither party may execute today. Each of the three is blocked by exactly one missing signed instrument.



One award in five states, and the four guards that must all hold at month nine. Two of the four are technical, one is regulatory, and the fourth belongs to an institution the company cannot bind.

3 The \$1.6M Gate and the \$3.5M Programme

3.1 One Budget, Reused Without Re-derivation

[DRAFTING INSTRUCTION] Open by stating that the five-year frame is taken verbatim and is not re-derived here: \$700,000 per year of direct cost, \$3,500,000 over five years, no cost share, no programme income anticipated, split across four layers of \$1,600,000, \$720,000, \$780,000 and \$400,000. Source: *funding/pdac-funding-applications/final-apply/sections/sec-08-budget-and-leverage.tex*. Cite *fundapp2*.

3.2 The Award, and What Sits Inside It

[DRAFTING INSTRUCTION] Explain the difference between an award and the work it buys. An NIH SBIR award is a total-cost figure: direct costs, indirect costs and fee. Set Table 8 with the three-component decomposition for both phases, ending at \$306,000 and \$1,300,000, and state the 15.0 percent overhead-and-fee load against the 40 percent indirect allowance the company does not claim. Cite *nihsbirseed* and *nihgpsfee*. Source: *funding/capitalization-plan/d2/fig-08-two-prices-one-programme.d2.md*.

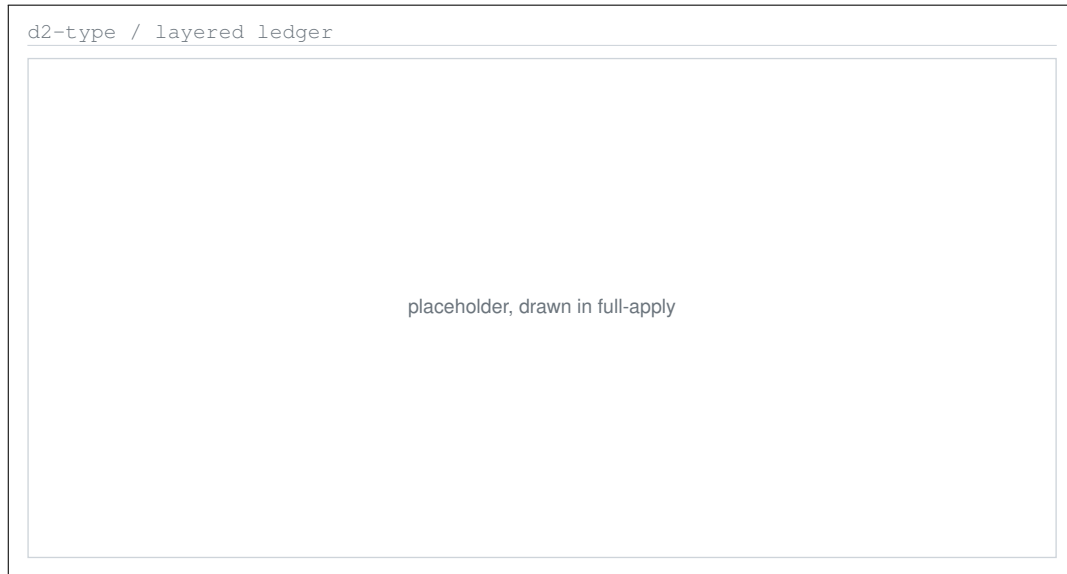
[DRAFTING INSTRUCTION] After Figure 7, one paragraph per guard, four short paragraphs. G4 gets the longest treatment, because it is the only guard the company cannot satisfy by working harder, and because the Descoped state exists only for it. Source: *funding/capitalization-plan/mermaid/fig-07-phase-gate-state-machine.md*.

3.3 Phase I, Nine Months, \$306,000

[DRAFTING INSTRUCTION] Set Table 9: eight line items summing to \$306,000 total cost. The rows are personnel for the CEO at 0.6 FTE and the verification engineer at 0.5 FTE, compute, rig hardware, regulatory, site fees, travel, and the indirect and fee lines. State what Phase I does not buy: no participant is consented and no drug is dispensed inside it.

3.4 Phase II, Twenty-Four Months, \$1,300,000

[DRAFTING INSTRUCTION] Set Table 10: ten line items summing to \$1,300,000 total cost, including six participants at \$48,000 each. State the descoped path: if the site executes late, Phase II is resubmitted for three participants rather than six, which is the Descoped state of Figure 7.



One programme, four layers, and two prices for the same work. The SBIR route reaches 1,396,000 of direct effort inside a 1,606,000 award; the third column is the 2,104,000 that it does not reach.



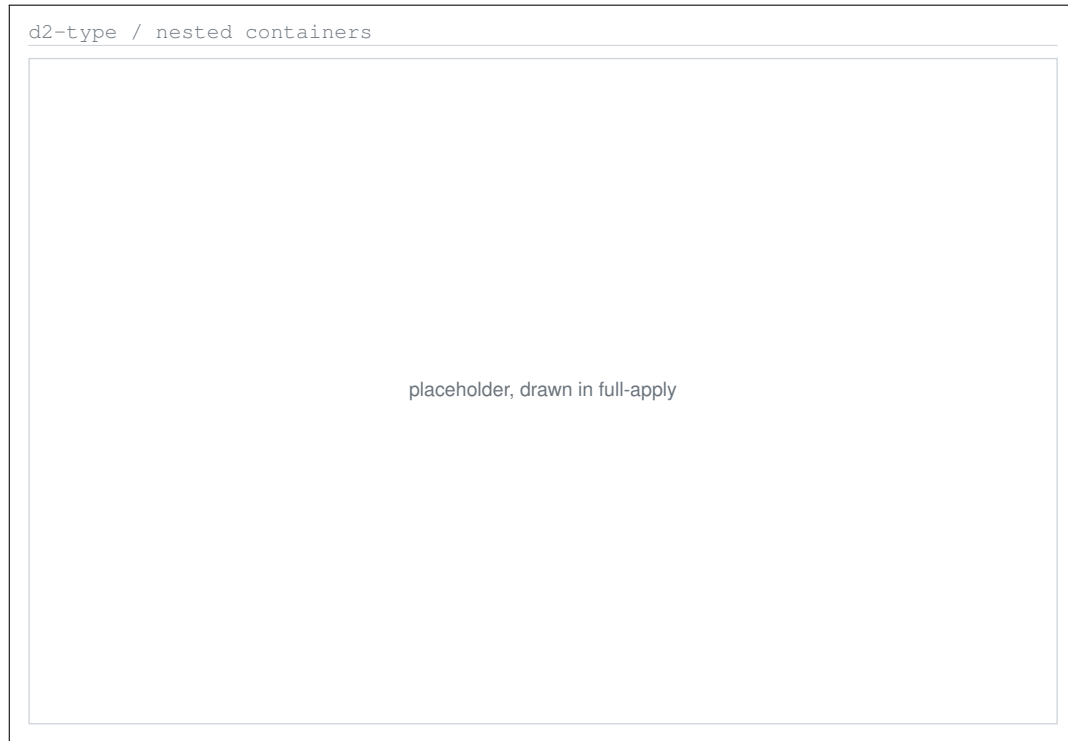
Seventeen work packages and every edge between them. Five close at 306,000, seven more at 1,606,000, and five are unreachable at that figure, all five downstream of the sixth participant.

3.5 The Same Work Priced Twice

[DRAFTING INSTRUCTION] After Figure 8, state the reconciliation in prose and give the four row-level checks. Then explain the delta in one paragraph: 27 months of additional term, participants 7 to 18, two more dose levels, the correlative pharmacology, the full VVUQ re-run, and the archive. Do not present the delta as a shortfall to be apologised for; present it as the honest price of the mechanism.

3.6 What the Money Reaches

[DRAFTING INSTRUCTION] After Figure 9, one paragraph on the structural finding: the money runs out one participant after the second dose level clears, and three of the five unreachable packages are downstream of that point. Then a short paragraph on WP15 and WP17, the two unreachable packages that are not about participants, and why WP17's removal changes what \$10 can claim. Source: *funding/capitalization-plan/graphviz/fig-09-work-package-dag.gv.md*.



Three capital tiers and the two gaps between them. Federal money at the base, contributed value unpriced in the middle, private capital above, and 5,900,000 against 1,606,000 is 3.67 to one.

4 Non-Dilutive to Dilutive Bridge

4.1 Three Tiers of Capital

[DRAFTING INSTRUCTION] Name the three tiers and their totals in the first paragraph: \$1,606,000 non-dilutive federal, contributed value that is deliberately unpriced, and \$5,900,000 dilutive private. Then state the leverage: 3.67 to 1 on cash alone, against the annex's at least 3 to 1 target, before any contributed value is counted. Source: `funding/science-golden-age/chunk-08-annex-fiscal-year-2028-research-and-development-budget-priorities.md`. Cite `kratsiosvought2026fy2028priorities`.

[DRAFTING INSTRUCTION] After Figure 11, one paragraph on why tier 2 carries no dollar figure, reusing the parent work's reasoning without restating it at length: no agreement exists, so a valuation would be invented, and an invented cost-share figure converts a real structural argument into an unverifiable number. Source: `funding/pdac-funding-applications/final-apply/sections/sec-08-budget-and-leverage.tex`.

4.2 The Capital Firewall

[DRAFTING INSTRUCTION] State the problem in two sentences before drawing it. The author is chief executive of the sponsor and holds 100 percent of it; every §54.2 trigger would fire at once if he were also a clinical investigator. The plan therefore assigns the investigator role entirely to the site. Cite `ecfr2026part54` and `fdafindisclosure2013`.

[DRAFTING INSTRUCTION] Set Table 11 after Figure 10: the five §54.2 and §54.4 triggers with their citation, threshold, and the form each requires. The five rows are in `funding/capitalization-plan/plantuml/fig-10-capital-firewall-guards.puml.md`. Then one paragraph on the disclosure period, which runs for the duration of the study and one year after completion, and why that term appears in the S5 to S3 return guard and nowhere else.



The capital firewall as five states. Two of them, below the rule, may not be entered while a participant is on study. Every guard names its own regulation and the form the entry would require.

4.3 What the Firewall Actually Separates

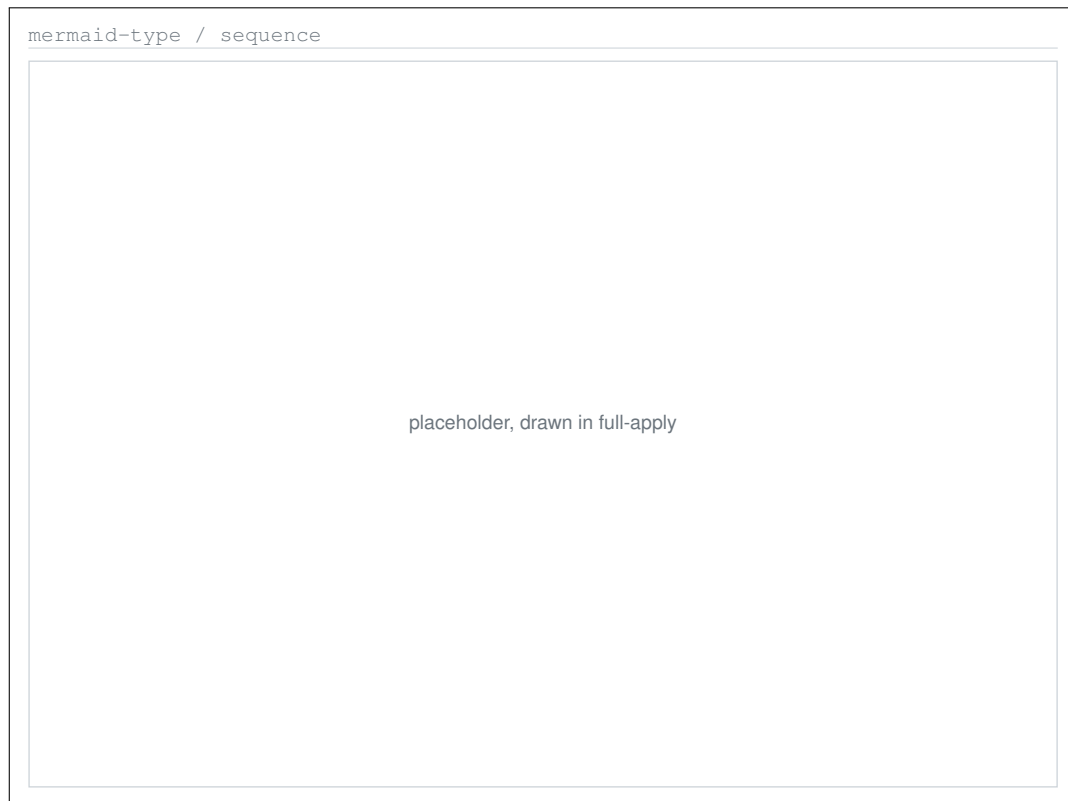
[DRAFTING INSTRUCTION] *Three short subsubsections or a three-row treatment, one each for dose assignment, endpoint adjudication and the analysis plan. Make the honest technical point explicitly: a Phase 1 3+3 escalation carries no randomization, so the firewall cannot bind on a randomization schedule; it binds on dose assignment, which is the Phase 1 analogue, and the wording is carried forward so it still binds when the Phase 2 successor introduces randomization. Source: `trial-protocol/` and `trial-phase-2/`. Cite `phase2proto`.*

4.4 The Capitalization Table

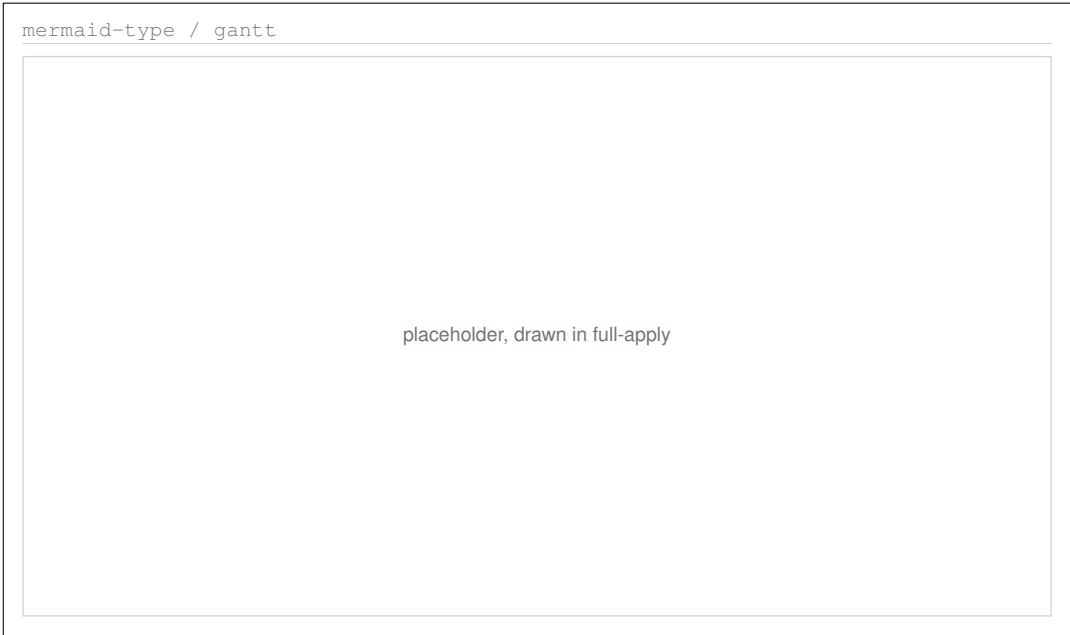
[DRAFTING INSTRUCTION] *Set Table 12: five stages, each with the instrument, the amount, the resulting founder percentage, and whether the `ecfr2026sbiownership` test still passes. Stage 5 must be shown failing, with the NIH venture-capital operating company authority named as the only route that would preserve eligibility, and with the plan's own answer, which is that stage 5 sits outside the 33 months this document covers.*

4.5 The Sequence

[DRAFTING INSTRUCTION] *After Figure 12, one paragraph on the two dates that matter: the 30-day notice to the site PI, and the update to the IND financial disclosure commitment at the next annual report. Close the section by naming the milestone each tier 3 instrument is gated behind, M8, M10 and M12, and stating that no private capital enters the entity before dose level 1 has cleared. Source: `funding/capitalization-plan/mermaid/fig-12-financing-event-sequence.md`.*



Eleven messages during a financing round, and the three lifelines no message reaches. Dose assignment, endpoint adjudication, and the analysis plan are untouched between term sheet and signature.



Thirty-three months, twelve milestones, and the artifact date on each. Phase I holds five at 306,000 dollars; Phase II holds seven at 1,300,000. The gate at month nine is the only hard boundary.

5 Twelve Milestones a Program Officer Can Audit

5.1 What an Auditable Milestone Is

[DRAFTING INSTRUCTION] Four sentences defining the contract this section offers. A milestone is auditable only if it carries four things: an evidence artifact that exists as a file, a cost that sums into its phase, a stop condition stated before the work begins, and a named funder mechanism that pays for it. State that all twelve rows carry all four, and that the two phase totals are exact.

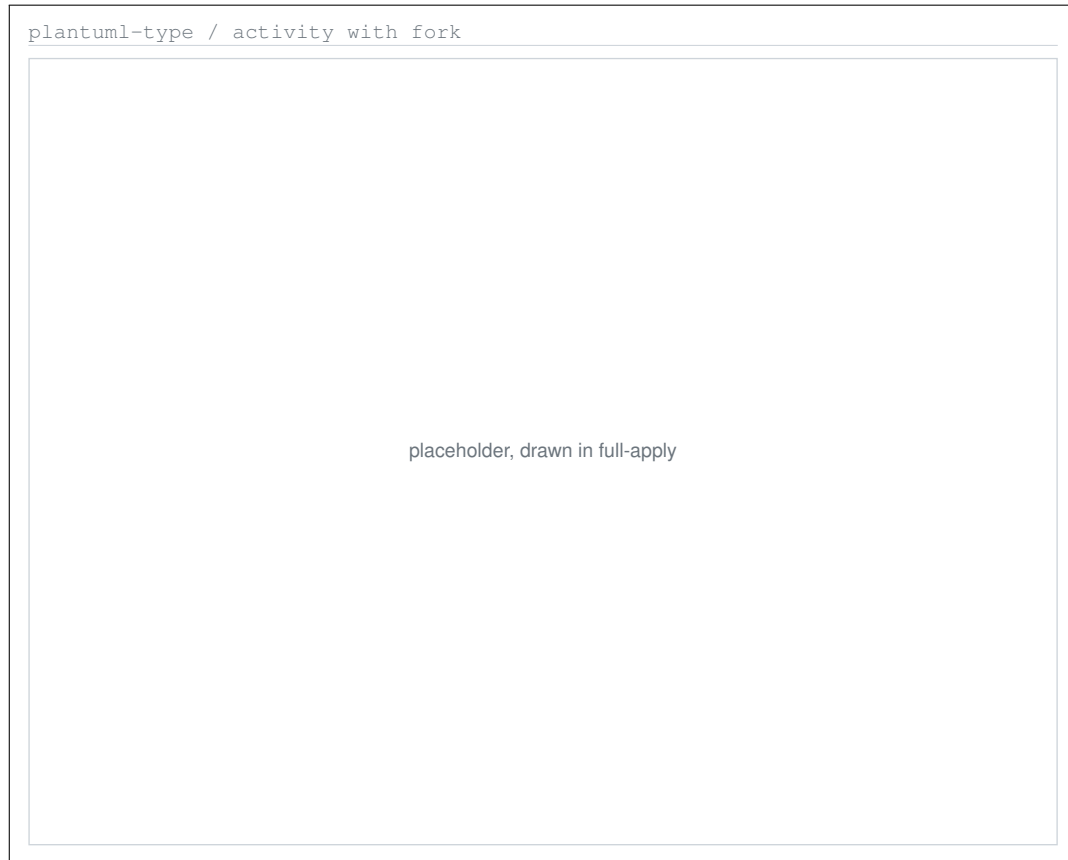
5.2 The Twelve Rows

[DRAFTING INSTRUCTION] Set Table 13, the central artifact of the paper. Six columns: ID, months, milestone, cost, evidence artifact, stop condition. Twelve rows. The first five must sum to \$306,000 and the last seven to \$1,300,000. Every row's data is in funding/capitalization-plan/mermaid/fig-13-twelve-milest one-calendar.md and every stop condition must be a number or a signed instrument, never a judgement. Because this table is wide, cut it as p{0.8cm} p{1.4cm} X p{1.6cm} for the first four columns and set the evidence artifact and stop condition as a second table, Table 14, keyed by the same twelve IDs, rather than compressing six columns into one measure.

[DRAFTING INSTRUCTION] Set Table 14: twelve rows keyed by milestone ID, three columns for evidence artifact, stop condition, and the funder mechanism that pays. The mechanism column takes four values only: SBIR Phase I, SBIR Phase II, the \$2,104,000 delta, and contributed. Source: funding/capitalization-plan/graphviz/fig-09-work-package-dag.gv.md for the mechanism assignment, which must agree with WP1 to WP12.

5.3 How a Milestone Is Closed

[DRAFTING INSTRUCTION] After Figure 15, one paragraph on why all three branches must join before an audit can begin: the hash proves the artifact has not moved, the source record proves it describes something that happened, and the monitoring report proves a third party looked at both. Any one alone is a claim, not evidence. Source: fund ing/capitalization-plan/plantuml/fig-15-milestone-evidence-activity.puml.md and funding/pdac-funding-applications/final-apply/sections/sec-06-physical-ai-gover nance.tex. Cite ecfr2026part312records.



*One milestone, three concurrent evidence branches, and the join
that has to close before a program officer can audit anything.
The replay is the only step that can send a milestone backwards.*

5.4 How the Programme Stops

[DRAFTING INSTRUCTION] After Figure 14, two paragraphs. The first names the two OR branches as the honest part of the figure: H1 and H5 are the only single points of failure and neither is technical. The second explains why the top event is a stop without a deposited archive rather than a stop, and walks the survival band: at every halt point the archive is already deposited for every closed milestone, so reaching the top event requires a halt plus a failure to deposit. Forward-reference Figure 20. Source: [funding/capitalization-plan/graphviz/fig-14-stop-condition-fault-tree.gv.md](#).



*One top event, five halt points, and ten basic events below.
Three are AND gates, so no single failure stops the programme.
The two OR branches are the month-nine gate and ownership test.*

6 Placeholder

7 Placeholder

8 Placeholder

9 Placeholder

10 Placeholder

11 Placeholder
